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Report on the Examination

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REPORT ON THE EXAMINATION: INTERNATIONAL – JANUARY 2026

This paper was well attempted by students. The mean mark was similar to previous series. There was a high standard deviation, showing that the paper differentiated students well.

As in most previous series, students performed better on average in the multiple-choice section C than in Sections A and B. In every section C question, the most popular response was the correct one.

Students gave some particularly good explanations in this paper, with some well-worded explanations in the topics of electricity, optical fibres and atomic energy levels. In other topics, such as in fluorescent lamps and stationary waves, the students did not seem so well-prepared. There was some weak answers in those topics even for questions which were recall (AO1 knowledge-based assessment objective).

Some general principles to note that apply in every series:

- When a question prompts the candidate to annotate a diagram or graph, this is usually required to attain full marks. In this paper this was required in questions 05.2 and 09.2
- If an explanation-type question is about a relationship between variables, it is often required to state the relationship as part of the explanation. Often, simply quoting the relevant equation is required as part of the full answer. In this paper, this was required in 06.1 and 11.4.
- When a line of best fit is drawn, it must be drawn such that a better line cannot be drawn with a translation or rotation of more than 1mm. This is often best achieved by having equal numbers of points above and below the line, on both sides of the centre of the line. The line must be a thin line (less than half a mm or small grid square in width), with no feathering or discontinuity. This is best achieved by using a sharp pencil with a 30cm ruler.

QUESTION A1

This 2-mark calculation was well answered, with over three quarters of students getting full marks. The most common error was using the number of slits per mm as the slit spacing.

QUESTION A2.1

This was well answered, with around 4 in 5 students scoring the mark. The next most popular option was As^{-1}.

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QUESTION A2.2

This was quite well answered, with around 3 in 4 students scoring the mark. A common error was converting from hours to seconds incorrectly.

QUESTION A3

This question was well answered, with around 2 in 3 students scoring the full 3 marks. A common error was using the equation $E = hc/\lambda$, treating the electrons as photons travelling at the speed of light. Some students chose the correct equations but rearranged them incorrectly.

QUESTION A4.1

This was well answered, with around 4 in 5 students scoring the mark.

QUESTION A4.2

This question was found to be one of the hardest in the paper. Under half of the students scored any marks. A common misconception seen was the idea that increasing the critical angle makes it easier (not harder) for light to undergo total internal reflection, resulting in less light leaving the fibre. In fact, the opposite is the case: fewer paths of light undergo total internal reflection, and light travelling with longer paths refracts out of the fibre, reducing modal dispersion.

QUESTION A5.1

This circuits calculation was well answered with around 3 in 4 students scoring the full 3 marks. Common errors included assuming the voltage across both components must be the same, or reversing the voltage across the lamp and the resistor.

QUESTION A5.2

Students found this question slightly harder and were more likely to score just 1 mark than both. Common errors included not showing a tangent on the graph as prompted, and rounding the upper limit to 1V rather than 0.90V.

QUESTION A6.1

Most students scored just 1 out of the 2 marks in this question. Many students did not refer to the equation. Other students used the term ‘area’ vaguely without ‘cross-sectional’.

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QUESTION A6.2

This longer calculation was well answered, with over half of the students scoring the full 4 marks. Some students found it challenging to distinguish between the different lengths and diameters in the question, leading to incorrect values for the length and cross-sectional area of the wire.

QUESTION A6.3

Around half of the students scored the mark for this question, which was an unusual context-based question. There were some well-worded responses, including some students correctly identifying that some of the long wire must still be in the circuit to enable terminal B to be connected. A few students suggested that the ohmmeter might have its own resistance but this did not score the mark as the question said that the ohmmeter was giving an accurate reading.

QUESTION A6.4

This was well answered, with over 4 in 5 students scoring full marks. Some students did not calculate the current correctly, choosing an incorrect combination of voltage and resistance.

QUESTION A7.1

This question yielded a spread of marks; most students scored at least 1 mark but only a third scored both. Some students confused the terms ‘ground state’, ‘excitation’ and ‘ionisation’, for example, suggesting that the energy at ‘the excitation level’ or at the ground state is defined as zero. Some students referred to the ‘surface’ of mercury, possibly confusing atomic energy levels with the photoelectric effect.

QUESTION A7.2

Just under two-thirds of students identified the correct transition.

QUESTION A7.3

This question was well answered, with over 60% of students scoring full marks. The most common error was students not converting from eV to J correctly.

QUESTION A7.4

This question yielded a spread of marks; most students scored at least 1 mark but less than half scored both. One common error was suggesting that the mercury atoms collide with each other to cause excitation. Many responses were too vaguely expressed, with the free-moving electrons being confused with the atomic electrons.

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QUESTION A7.5

This question yielded a spread of marks, with approximately equal numbers scoring 0,1,2,3 marks. We required students to refer to photons to receive full marks for this question because de-excitation is quantised. A misconception that was seen was that the mercury atoms emit (primarily) visible light, and those students did not refer to the phosphorescent coating.

QUESTION A8.1

This question was well answered with 3 in 5 students scoring full marks. Some students mixed up the different lengths in the question, for example using the slit width instead of slit spacing.

QUESTION A8.2

This question was intended to be relatively challenging, and around 1 in 3 students scored the mark. A common suggestion was that the diffraction pattern does not arise at all, because the slit width is larger than the wavelength, which is incorrect. (For comparison, in the more familiar visible light Young's slit setup, the slit width is usually larger than the wavelength.)

QUESTION A8.3

This question was well answered with 7 in 10 students scoring the mark. We did not require knowledge of Malus's Law, but simply two maxima and minima per 360°.

QUESTION A8.4

Students found this question harder, with just under half scoring at least one mark. Many students were vague about the need to align the aerial with the oscillations of the wave, simply stating that the aerial 'needs to be vertical' with no further explanation.

QUESTION A8.5

Most students scored at least one mark in this question. The majority identified that the graph is a flat line but many put it at the level of a maximum from 8.3.

QUESTION A9.1

This question was well answered and over 8 in 10 students scored the mark. Most referred to air resistance.

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QUESTION A9.2

Most students scored at least one mark in this question. However, many students referred to the equilibrium point in their written answer without annotating the diagram. If annotation is prompted in the question, then this is often required to gain full marks, as it was here.

QUESTION A9.3

This question was not well answered, with under half scoring any marks. Many students did not distinguish between absolute uncertainty (which is based on the reaction time and cannot easily be reduced) and percentage or relative uncertainty, which decreases proportionately when multiple periods are measured.

QUESTION A9.4

This question was intended to be challenging but students did well, with over half scoring full marks. Some expected the value of g obtained to be lower than 9.81 ms^{-1} instead of higher.

QUESTION B10.1

Around 3 in 5 students scored full marks for this question. Almost all students calculated the table values correctly with correct significant figures, but many plotted the points incorrectly.

QUESTION B10.2

Around half of students scored the mark for their line of best fit. As outlined above, the general tips for a line of best fit are as follows: it must be drawn such that a better line cannot be drawn with a translation or rotation of more than 1mm. This is often best achieved by having equal numbers of points above and below the line, on both sides of the centre of the line. The line must be a thin line (less than half a mm or small grid square in width), with no feathering or discontinuity. This is best achieved by using a sharp pencil with a 30cm ruler.

QUESTION B10.3

The majority of students scored 1 out of 2 marks for this question. Many rearranged to make R the subject without going on to compare directly to $y = m\,x + c$.

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QUESTION B10.4

Around a third of students scored full marks for this question. The final two marks were 'quality marks', meaning that we required an answer in the stated range regardless of students' answers to previous parts of the question.

QUESTION B11.1

Most students scored 1 or 2 marks for this question. Some suggested that the two progressive waves are coherent; this was not allowed as they are not spatially coherent. Many students did not refer to superposition.

QUESTION B11.2

This question was well answered, with 3 in 5 students scoring the mark. The two common errors were not converting μ correctly into SI units, and rearranging the equation incorrectly.

QUESTION B11.3

This was well answered with around 3 in 5 students scoring the mark. A common misconception was expecting the antinode in the centre, where there would be a node.

QUESTION B11.4

This question was the last in the written part of the paper so it was intended to be challenging. Many students identified that the tension increases but were vague that it 'increases more than the length'; we required the sense of the increase being by a larger factor or ratio. Around half of students scored 1 mark but very few scored both.

QUESTION C12

This was well answered, with around 3 in 4 students scoring the mark.

QUESTION C13

This was found to be slightly harder, with around 3 in 5 students scoring the mark. C was the second-most popular choice.

QUESTION C14

This was well answered, with around 4 in 5 students scoring the mark.

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QUESTION C15

This was quite well answered, with around 2 in 3 students scoring the mark. C was the second-most popular choice.

QUESTION C16

This was found to be slightly harder, with around 3 in 5 students scoring the mark. A and C were the next most popular choices.

QUESTION C17

This was well answered, with around 3 in 4 students scoring the mark.

QUESTION C18

This was well answered, with around 4 in 5 students scoring the mark.

QUESTION C19

This was well answered, with around 3 in 4 students scoring the mark.

QUESTION C20

This was well answered, with around 3 in 4 students scoring the mark.

QUESTION C21

This was found to be slightly harder, with around 3 in 5 students scoring the mark. C and D were the next most popular choices.

QUESTION C22

This was quite well answered, with around 2 in 3 students scoring the mark.

QUESTION C23

This was well answered, with around 4 in 5 students scoring the mark.

QUESTION C24

This was quite well answered, with around 2 in 3 students scoring the mark. B and C were the next most popular choices.

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QUESTION C25

This was well answered, with around 3 in 4 students scoring the mark.

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